Marketing Analytics

Marketing Decisions and AI-Assisted Coding

By Nathan Yang

Session 01



A Renewal Offer

- A subscription firm can contact 5,000 eligible customers.
- The decision concerns incremental contribution over the next 30 days.
- Past spending, consent, and account history are available before contact.

Analytical Questions

- Description: how often do customers renew?
- Prediction: who is likely to renew?
- Causal effect: how does an offer change renewal?
- Decision: which contacts generate positive incremental contribution?

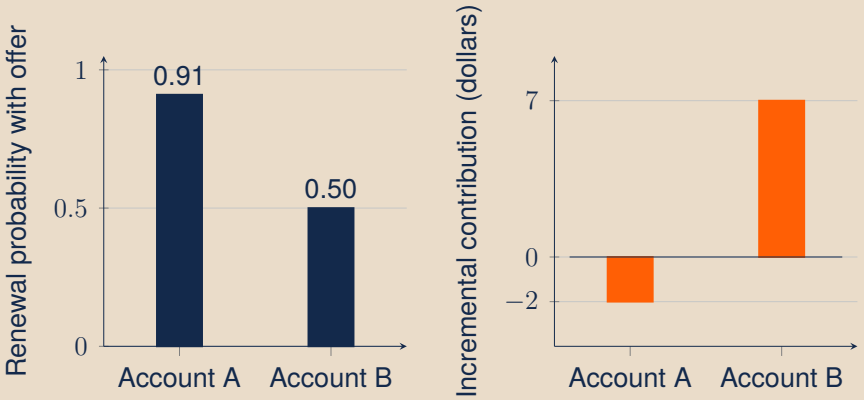
Incremental Contribution

For account i, compare renewal under an offer and no offer.

$$\Delta\pi_i = m_i\{p_i(1) - p_i(0)\} - c_i$$

Contribution m_i and contact cost c_i are dollars per account over the stated horizon.

Response and Incremental Value



Synthetic accounts. Contribution is 100 dollars and contact cost is 3 dollars.

The Observation Unit

- A product line, order, customer, and customer-month are different units.
- The key identifies one row at the intended level.
- Aggregation changes the weights used in an average.

Python Objects and Order Totals

Three synthetic product lines belong to two orders.

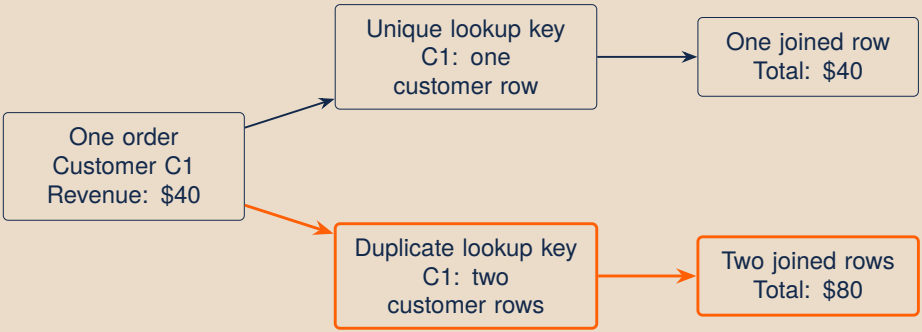
```
lines = [('A', 2, 10), ('A', 1, 5), ('B', 1, 12)]
orders = {}
for order, quantity, price in lines:
    orders[order] = orders.get(order, 0) + quantity * price
total = sum(orders.values())
aov = total / len(orders)
assert (total, aov) == (37, 18.5)
```

Check: total sales are 37 dollars and average order value is 18.50 dollars.

Functions and Data Contracts

- Inputs have names, types, units, and allowed missing values.
- A function returns a defined output and an exception report.
- Returns, canceled orders, and duplicate lines need explicit rules.

A Duplicate Lookup Doubles Sales



A valid-looking join can double revenue.
Check lookup-key uniqueness before merging, then reconcile totals.

Synthetic join. One 40-dollar order matches two customer records.

An AI Coding Request

- Return one row per order and an exceptions table.
- Treat negative quantities as returns under the stated rule.
- Verify the 37-dollar example and reject missing order IDs.
- Explain each transformation before adding the next step.

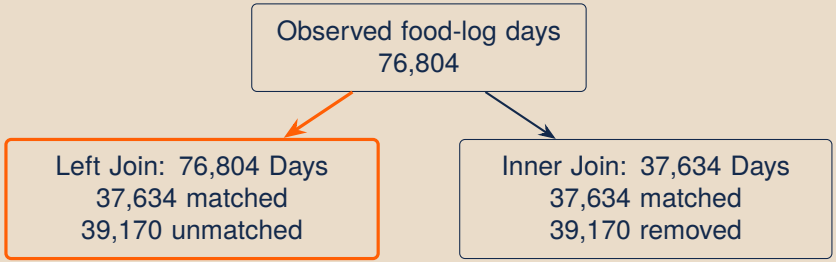
Verification With Counterexamples

- A duplicate lookup must trigger a failed key check.
- An unmatched customer must remain visible in the audit.
- An empty input needs an explicit result.
- Order totals must reconcile with eligible product lines.

mHealth Records

- Food entries describe recorded items on observed user-days.
- The separate daily table covers only part of the food-log history.
- Reported daily calories repeat across item rows.

Join Choice Changes the Sample



Same source keys; different analysis populations.
An unmatched day is still an observed food-log day.

Released mHealth cohort: 76,804 observed user-days.

Coding Exercise: An Order Audit

- Implement aggregation with a coding assistant.
- Add a return, a missing identifier, and a duplicated lookup key.
- Report how each change affects rows, totals, and exceptions.

A Reproducible Analysis

- Relative paths identify inputs and outputs.
- A fresh run reconstructs results without hidden notebook state.
- Record package versions and consequential AI contributions.

A Campaign Specification

- Define the eligible customers and contact date.
- State the contribution horizon and relevant costs.
- Identify the evidence needed for the untreated outcome.
- Choose a numerical check that would reveal a wrong denominator.

Customer Data in an AI Workflow

- Use only data authorized for the tool and task.
- Minimize fields and use synthetic examples when they suffice.
- Check where uploads, logs, and generated files are stored.

Decision Discussion

- Which customer receives the renewal offer?
- What changes if every treated renewal receives a discount?
- Which future variable would invalidate a Monday-morning forecast?

Marketing Analytics

Customer Data and Measurement

By Nathan Yang

Session 02

A Customer-Activity Decision

- An mHealth manager wants to understand reduced recorded use.
- The comparison follows the same eligible customers over time.
- A missing record can reflect nonuse, incomplete observation, or data coverage.

A Customer Table

- Begin with the eligible-customer roster.
- Aggregate events to the customer-period level.
- Left join the summaries to the roster.
- Fill zero only when absence has the defined meaning of no recorded event.

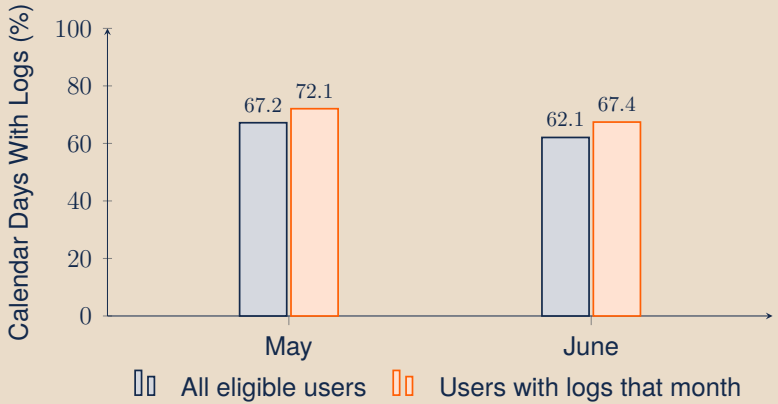
Two Activity Measures

Let D_i be observed logging days and T the calendar days in a common window.

$$L_i = \frac{D_i}{T}, \qquad \bar{L} = \frac{1}{N}\sum_{i=1}^{N} L_i$$

Each eligible customer receives equal weight, including customers with no recorded days.

Conditional and Fixed Denominators



mHealth activity depends on who remains in the denominator.

Repeated Totals

- An item row can contain a copied daily total.
- Five copies of 1,132.79 sum to 5,663.95.
- The item-calorie sum on that example day is 1,132.7852.

Structural Errors and Review Flags

- A duplicate daily key violates a one-row-per-day contract.
- Equal item descriptions can represent repeated servings.
- An extreme value warrants review under a stated rule.

A Fixed Customer Denominator

Synthetic counts follow all four eligible customers.

```
days = {'A': 20, 'B': 10, 'C': 0, 'D': 0}
fixed = sum(days.values()) / (4 * 30)
active = [x for x in days.values() if x > 0]
conditional = sum(active) / (len(active) * 30)
assert round(fixed, 4) == 0.25
assert round(conditional, 4) == 0.50
```

The fixed-roster rate is 25%. The active-user rate is 50%.

Cohorts and Customer Age

- Group customers by a defined entry period.
- Compare the same time since entry.
- Mark later outcomes as unobserved when the horizon has not elapsed.

Retention Has a Triangular Support

	At Entry	Month 1	Month 2	Month 3
January	100%	60%	45%	40%
February	100%	70%	50%	Not yet observed
March	100%	80%	Not yet observed	Not yet observed

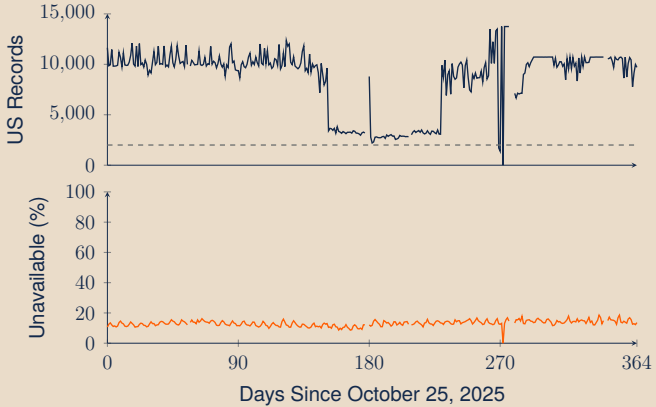
Each cohort begins with 100 customers.
Only observed customer ages enter the retention denominator.

Synthetic cohort structure. Blank cells have no elapsed follow-up.

McBroken as a Measurement Case

- The records describe ice-cream ordering availability.
- A missing snapshot has an unknown status.
- Restaurant coverage and status recognition can change over time.

Coverage Changes the Percentage



McBroken comparisons preserve the observed-store denominator.

Identity Across Files

- A restaurant address is a candidate key that requires checks.
- Standardizing spelling can repair formatting differences.
- Overaggressive matching can merge separate locations.

Coding Exercise: A Data Audit

- Construct a roster-first activity table.
- Report key uniqueness, unmatched rows, and zero-event customers.
- Compare the fixed and conditional activity measures.

AI Review of a Parser

- Supply the allowed statuses and a tiny labeled example.
- Require an explicit unknown category for new statuses.
- Compare parsed totals with an independently calculated source total.

A Data Dictionary for Decisions

- State the unit, key, timing, and units of each variable.
- Record source versions and transformation rules.
- Separate observed values, calculated quantities, and imposed assumptions.

Privacy and Collection Choices

- Collect the fields needed for the stated decision.
- Retain source permissions and deletion or retention conditions.
- Check whether a linkage increases the risk of identifying a person.

Measurement Discussion

- Could recorded activity rise because low-activity users disappear?
- Would deleting unmatched days change the target population?
- What evidence would establish a mechanical failure?

Marketing Analytics

Visualization and Data Quality

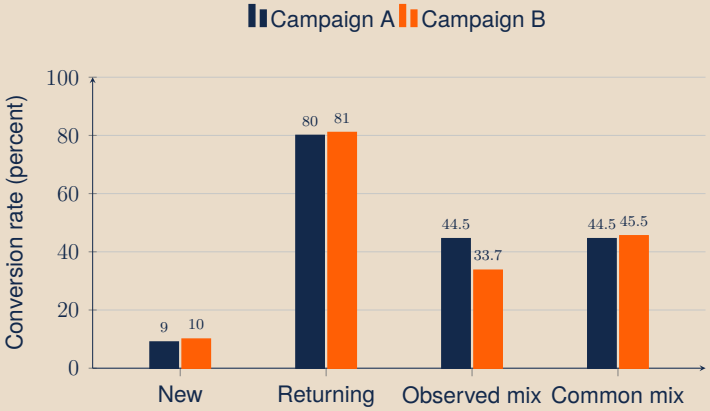
By Nathan Yang

Session 03

A Campaign Comparison

- Campaign B converts more new and returning customers within each group.
- Its aggregate conversion rate is lower.
- The customer mix determines which comparison the aggregate makes.

Customer Mix Reverses the Ranking



Synthetic campaigns. Both within-group rates rise by one percentage point.

A Standardized Conversion Rate

Use common customer-group weights w_g for each campaign c.

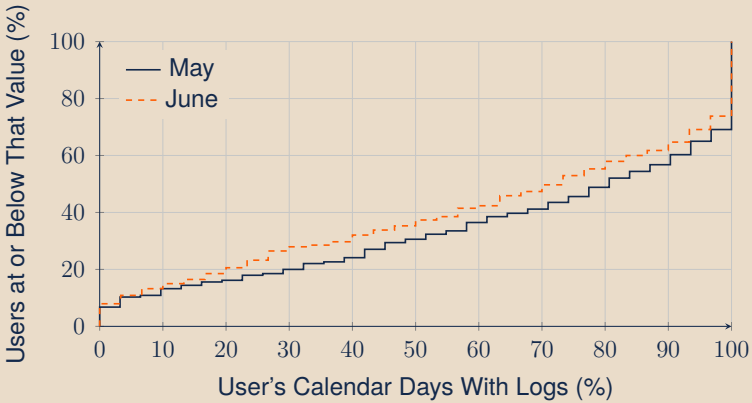
$$\bar{p}_c^{\text{std}} = \sum_g w_g p_{cg}, \qquad \sum_g w_g = 1$$

Changing the weights changes the population represented by the rate.

Chart Choice Follows the Question

- Aligned positions compare magnitudes.
- Lines compare changes over ordered time.
- Distributions reveal spread and concentration.
- Paired displays preserve within-customer changes.

A Mean Decline Can Hide a Stable Median



mHealth comparison uses paired customers and a common observation window.

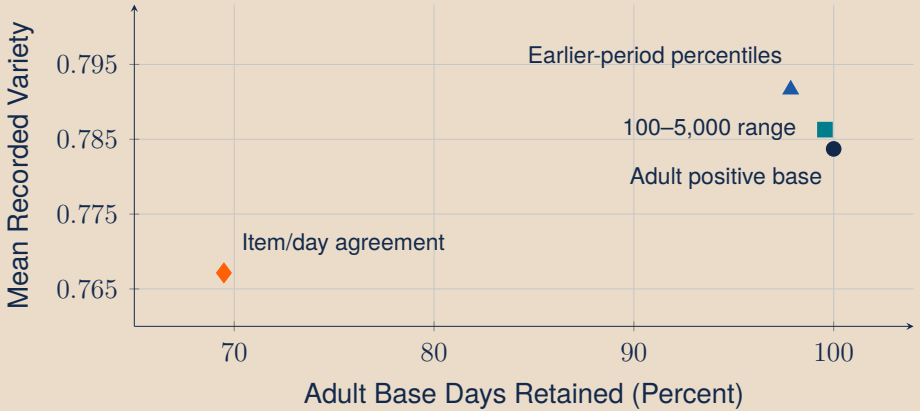
Uncertainty and Dependence

- Repeated days for one customer share common influences.
- Customer-level resampling preserves each customer history.
- A confidence interval and a percentile range describe different objects.

Nutrition Knowledge and Measurement

- A recorded food log can omit meals and portions.
- Energy plausibility depends on the measurement instrument and physiology.
- Age, body size, activity, and recording duration affect screening.

Sample Restrictions Change the Comparison



mHealth screening alternatives are explicit analysis choices.

Energy-Plausibility Screens

- An intake-to-requirement comparison needs defined inputs.
- Unobserved physiological inputs remain missing.
- A hypothetical Goldberg calculation teaches the method separately from the logs.

Interactive Comparisons

- Keep comparable panels on common scales.
- Display the eligible population and current selection.
- Preserve customers with no recorded days when the estimand includes them.

A Composition Check

Synthetic within-group rates are 9%, 80%, 10%, and 81%.

```
rates_a = [0.09, 0.80]
rates_b = [0.10, 0.81]
common_a = sum(rates_a) / 2
common_b = sum(rates_b) / 2
observed_b = (2 * rates_b[0] + rates_b[1]) / 3
assert round(common_b - common_a, 2) == 0.01
```

B improves by one point under common weights despite its lower observed aggregate.

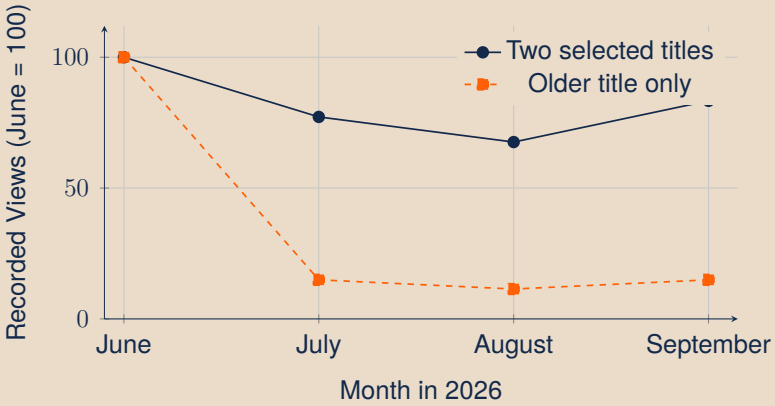
Coding Exercise: A Chart Audit

- Generate a rate chart from a verified aggregate table.
- Alter the axis range, cohort filter, and denominator one at a time.
- Explain which changes alter the substantive comparison.

A Public-Signal Measurement Problem

- A product can acquire a new page title while demand stays unchanged.
- A single-title series then misses part of its recorded attention.
- Coverage, redirects, and missing dates require an audit.

Title Changes and Apparent Declines



The supplied public pageview capture uses observed title histories.

Evidence in a Management Presentation

- State the decision and supported result.
- Show the comparison, units, and denominator.
- Keep the qualification that could change the action beside the claim.

Honest Comparisons

- Keep denominators and scales visible.
- Preserve missing outcomes and relevant subgroup variation.
- Avoid presenting a filtered descriptive pattern as a general causal effect.

Visualization Discussion

- Which customers receive weight in the chart?
- Could sample restrictions reverse the recommendation?
- What does the uncertainty interval include?

Marketing Analytics

Regression and Marketing Interpretation

By Nathan Yang

Session 04

Competitors and Store Sales

- Stores in markets with more competitors can have higher sales.
- Demand attracts firms and also raises sales.
- The managerial question concerns the effect of an additional competitor.

A Conditional Mean

Relate store sales to observed marketing conditions.

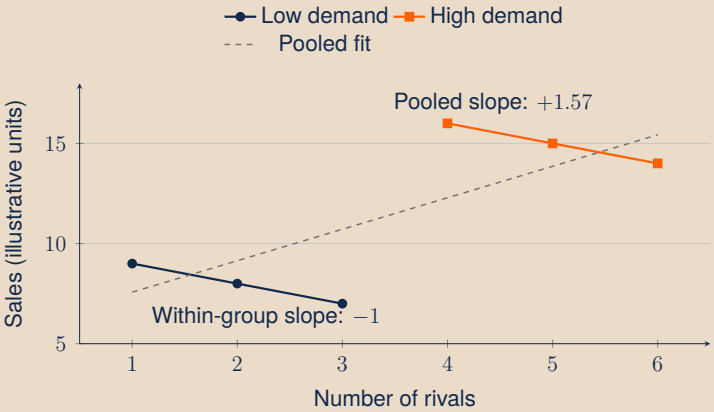
$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 Z_i + u_i$$

The coefficient on X_i describes its linear association holding included Z_i fixed.

Ordinary Least Squares

- OLS chooses coefficients to minimize squared residuals.
- A residual is observed sales minus fitted sales.
- The model summarizes the comparison supplied by the data.

Omitted Demand Reverses the Sign



Synthetic markets hold the underlying competitive mechanism fixed.

Omitted-Variable Bias

Suppose demand Z affects sales and is omitted from a regression on X.

$$\widetilde{\beta}_1 = \beta_1 + \beta_2 \frac{\mathrm{Cov}(X, Z)}{\mathrm{Var}(X)}$$

The omitted association can offset or reverse the coefficient of interest.

Price and Sales Units

- A level coefficient is sales units per dollar.
- A log-log price coefficient is an elasticity under the model.
- The exact change for a log outcome uses an exponential transformation.

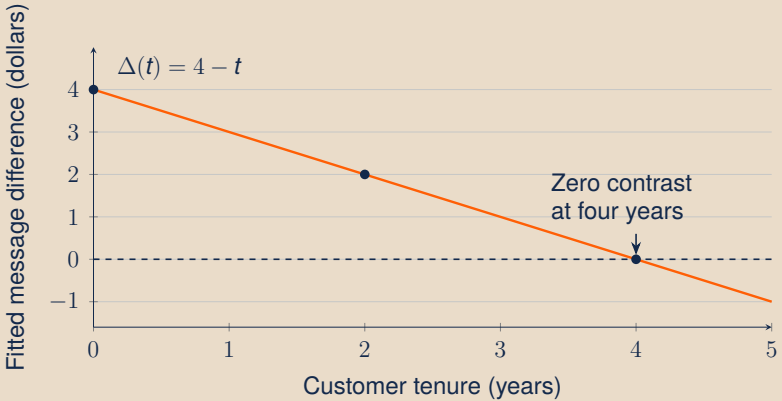
Interactions

Allow a message association to vary with account tenure T.

$$Y = \beta_0 + \beta_1 M + \beta_2 T + \beta_3 MT + u$$

The message comparison at tenure T is $\beta_1 + \beta_3 T$.

A Comparison Changes With Tenure



Synthetic message-response model. Tenure determines the conditional comparison.

Uncertainty and Practical Importance

- Report effect size in managerial units.
- Cluster at the unit that shares outcome disturbances or assignment.
- Compare the interval with a decision-relevant magnitude.

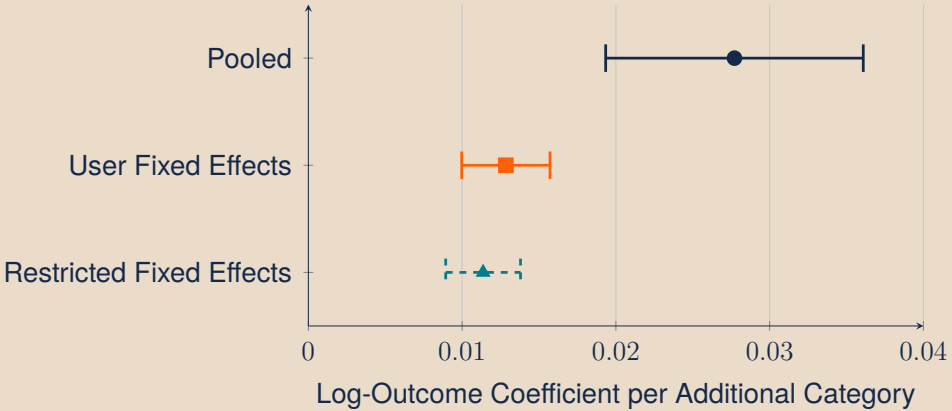
Panel Comparisons

- Customer effects absorb stable customer characteristics.
- Time effects absorb shocks shared within a period.
- Time-varying confounding can remain.

mHealth: The Next Calendar Day

- Relate recorded category breadth to next-day recorded calories.
- Match exact adjacent calendar dates.
- State the included users, days, and calorie restrictions.

Screening and the Estimated Association



mHealth estimates retain the stated outcome timing and sample restrictions.

A Hand-Checkable Regression

Synthetic sales follow a linear relationship.

```
x = [1, 2, 3, 4]
y = [3, 5, 7, 9]
mx, my = sum(x) / 4, sum(y) / 4
b = sum((a-mx)*(v-my) for a,v in zip(x,y))
b /= sum((a-mx)**2 for a in x)
a = my - b * mx
assert (a, b) == (1.0, 2.0)
```

The intercept is 1 and the slope is 2 sales units per unit of x.

Coding Exercise: Regression Checks

- Generate a model after defining the outcome and information set.
- Verify the coefficient units and one fitted value manually.
- Compare pooled and customer-effect specifications.
- Inspect residuals, outliers, and sample changes.

Prices Respond to Demand

- Managers can raise prices when demand is unusually strong.
- An observed price-sales regression can mix demand and pricing choices.
- An instrument needs relevance and a defensible exclusion restriction.

Evidence for an Advertised Claim

- A claim that a product cuts costs needs a suitable comparison.
- An association can reflect selection and omitted demand.
- The evidence should support the express and implied claim before publication.

Regression Discussion

- Which comparison supports the coefficient?
- What variable could reverse its interpretation?
- Would the recommended action change across the reported interval?

Marketing Analytics

Conjoint and Product Design

By Nathan Yang

Session 05



A Subscription-Bundle Decision

- A firm can change quality, delivery, and monthly price.
- Customers must trade off feasible attribute combinations.
- The choice set includes the option to buy nothing.

Choice-Task Design

- Vary price independently of features where feasible.
- Use realistic combinations and clear descriptions.
- Retain respondent identity across repeated tasks.

Utility and Price

Systematic utility combines product features and price.

$$V_{ij} = \beta_q q_j + \beta_f f_j + \beta_p p_j$$

Utility has an arbitrary scale. Price is measured in dollars per billing period.

Logit Choice Probabilities

Normalize probabilities over the alternatives available to customer i.

$$P_{ij} = \frac{\exp(V_{ij})}{\sum_{k \in \mathcal{J}_i} \exp(V_{ik})}$$

The denominator includes the outside option and every available product.

Willingness to Pay

Under linear price utility, equate a feature gain with a price increase.

$$\mathrm{WTP}_f = -\frac{\beta_f}{\beta_p}$$

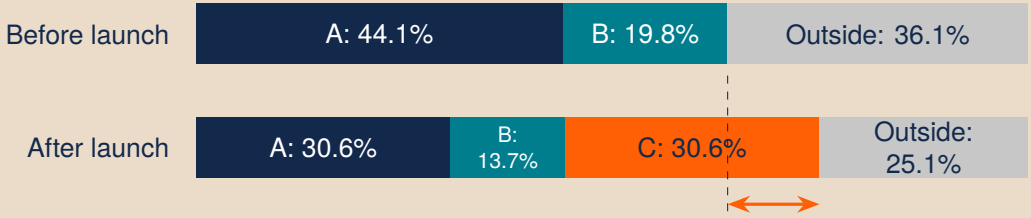
If $\beta_f = 1.2$ and $\beta_p = -0.1$ per dollar, the tradeoff is 12 dollars.

Two Bundles and an Outside Option

Option	Quality	Price	Utility
A	Premium	10	0.2
B	Standard	6	-0.6
Outside	None	0	0

Synthetic inputs: quality coefficient 1.2 and price coefficient -0.1.

A Launch Redistributes Existing Sales



Synthetic logit model. New bundle C has A's modeled attributes.

Incremental Contribution

- Attach contribution margins to each simulated purchase.
- Compare the full product portfolio before and after launch.
- Deduct development and operating costs separately.

A Probability Check

Calculate the synthetic baseline shares.

```
from math import exp
utilities = {'A': 0.2, 'B': -0.6, 'Outside': 0.0}
weights = {k: exp(v) for k,v in utilities.items()}
shares = {k: v/sum(weights.values())
          for k,v in weights.items()}
assert abs(sum(shares.values()) - 1) < 1e-12
```

A has about 44.1% share, B 19.8%, and the outside option 36.1%.

Substitution Assumptions

- A close copy can draw mostly from its nearest competitor.
- Basic logit instead preserves relative odds among existing alternatives.
- Holdout choice tasks can reveal systematic substitution errors.

Coding Exercise: A Product Launch

- Add C to the synthetic profile table.
- Calculate portfolio share and contribution before and after launch.
- Compare C’s share with the change in category participation.
- Check a case with no outside option.

Validation and Market Transfer

- Hold out respondents or tasks according to the intended use.
- Check coding, price units, and probabilities.
- Relate stated preference to awareness, distribution, and availability.

mHealth Product Economics

- Food logs reveal recorded use and category histories.
- A paid-plan choice model needs additional price and preference evidence.
- Financial scenarios can state the missing inputs explicitly.

AI-Assisted Conjoint Analysis

- Provide the choice-set identifier and the selected alternative.
- Require probabilities to sum to one within each task.
- Preserve respondent groups in validation.
- Request a portfolio comparison with explicit margins.

Price Presentation and Customer Understanding

- A base price and a mandatory fee determine the total paid.
- A stated-choice task can reveal different responses to equivalent totals.
- The feasible pricing design depends on the applicable rule and industry.

Product-Design Discussion

- How much of C’s demand comes from existing products?
- Which design change would identify the price tradeoff?
- What evidence could change the launch recommendation?

Marketing Analytics

Segmentation and Customer Lifetime Value

By Nathan Yang

Session 06

A Retention-Support Decision

- A service team can support only some customers.
- High baseline value and responsiveness to support are separate quantities.
- Segments should correspond to feasible service decisions.

Customer Features at a Common Date

- Recency is time since the last eligible event.
- Frequency counts events in a fixed window.
- Monetary value uses sales or contribution over that window.

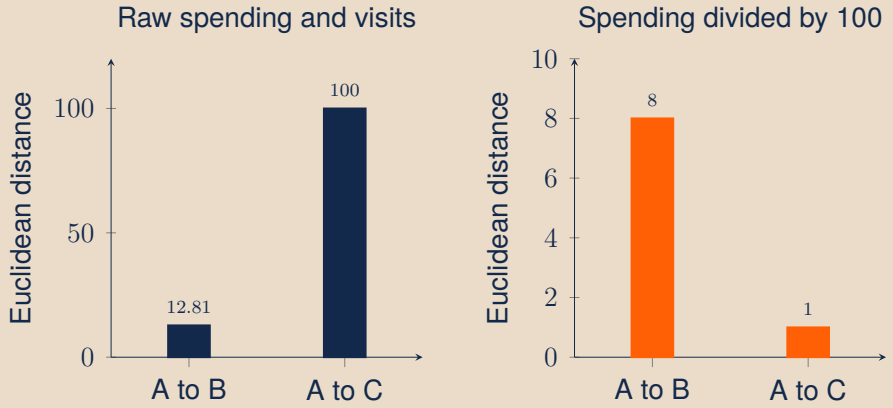
K-Means and Similarity

Choose K centers for transformed customer features z_i.

$$\min_{\mu, c} \sum_i \left\| z_i - \mu_{c(i)} \right\|^2$$

Scaling the features changes the distance and can change the segments.

Units Change the Nearest Customer

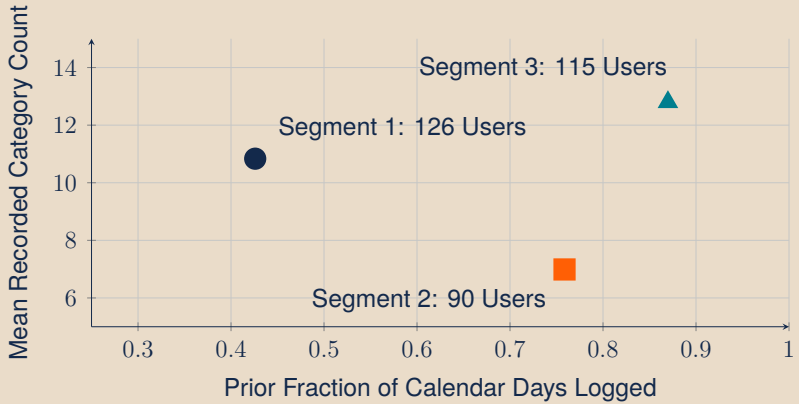


Synthetic customers have unchanged spending and frequency records.

Evaluating a Segmentation

- Inspect segment size, profile, and actionable differences.
- Compare stability across seeds, samples, and periods.
- Evaluate outcomes that did not mechanically define the clusters.

mHealth Segments Use Earlier Histories



Eligible users have at least 30 observed days before July 1.

Later Logging by Earlier Segment

Segment	Users	Prior Fraction	Later Fraction
1	126	0.426	0.338
2	90	0.759	0.594
3	115	0.870	0.654

mHealth denominators use 182 earlier and 184 later calendar days.

Lifetime Value and Payment Timing

Value future contribution conditional on remaining active.

$$\mathrm{CLV} = \sum_{t=1}^{T} \frac{\Pr(\text{active at } t) m_t}{(1+d)^t} - a$$

The period for margin m_t, discounting d, and retention must agree.

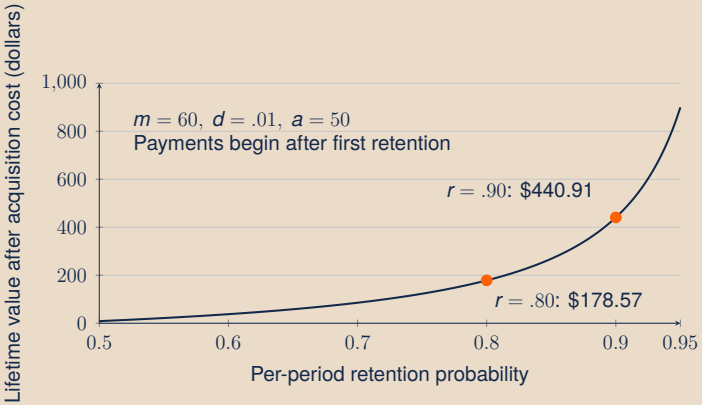
A Stationary Subscription

Suppose monthly retention is r and monthly contribution is m.

$$\mathrm{CLV} = \frac{mr}{1 + d - r} - a$$

With $m = 60$, $d = 0.01$, $a = 50$, and $r = 0.80$, CLV is 178.57 dollars.

Retention Has Nonlinear Value



Synthetic stationary model. Monthly margin, discounting, and acquisition cost stay fixed.

A Finite-Sum Check

Compare the closed form with 2,000 future monthly payments.

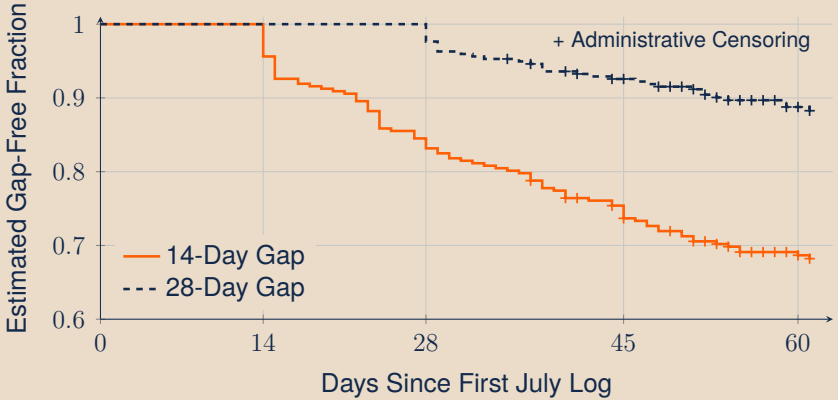
```
m, r, d, a = 60, 0.8, 0.01, 50
closed = m * r / (1 + d - r) - a
finite = sum(m * r**t / (1+d)**t
             for t in range(1, 2001)) - a
assert abs(closed - finite) < 1e-8
```

Both calculations give approximately 178.57 dollars.

Cancellation and a Logging Gap

- A subscription may record cancellation directly.
- A 30-day logging gap is an operational activity event.
- Observation can end before a defined event occurs.

The Event Definition Changes Survival



mHealth time-to-gap analysis uses a stated gap definition and observation horizon.

Endpoint and Continuous Retention

- Endpoint retention allows customers to leave and return.
- Continuous retention requires uninterrupted presence.
- Acquisition can include first entry, return, or switching.

Coding Exercise: Segments and Value

- Fit feature scaling and segments on an earlier period.
- Describe a later period without refitting the groups.
- Vary assumed paid retention in a separate CLV calculation.

Retention and Effective Cancellation

- High paid retention can reflect satisfaction or difficulty leaving.
- Measure cancellation completion and unwanted subsequent charges.
- Evaluate customer value alongside the seller’s lifetime-value objective.

Retention Discussion

- What action differs across the proposed segments?
- Does a customer’s absence represent churn or censoring?
- Which retention assumption changes the support decision?

Marketing Analytics

Text, Social Media, and Networks

By Nathan Yang

Session 07

A Customer-Service Decision

- A retailer wants to detect delivery complaints quickly.
- The document may be a review, sentence, or service conversation.
- The action is a review queue with limited staff capacity.

Text Preparation

- Keep negation and domain-specific terms when they affect meaning.
- Document language, duplicates, and boilerplate.
- Fit the vocabulary on training documents.

TF-IDF Representation

Weight a term by frequency in a document and rarity across documents.

$$\mathrm{tfidf}_{td} = \mathrm{tf}_{td} \log\left(\frac{N}{\mathrm{df}_t}\right)$$

Tokenization, smoothing, and normalization are implementation choices that need a record.

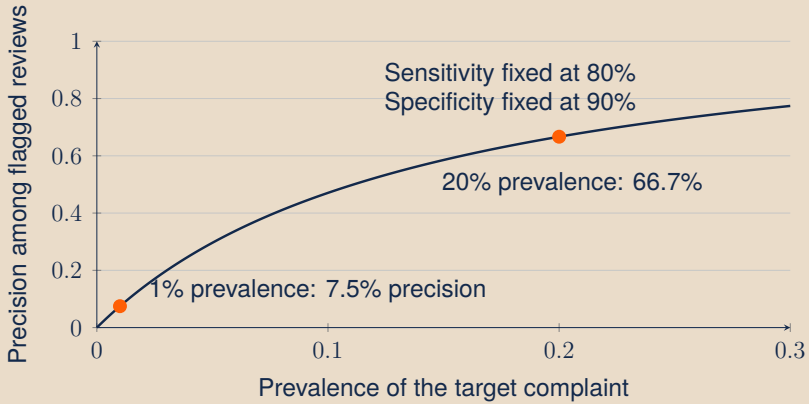
Embeddings and Similarity

- Embeddings place text in a numerical representation.
- Nearby descriptions may share meaning or usage.
- A useful similarity measure still requires domain validation.

Labels, Sentiment, and Topics

- Sentiment measures an evaluative construct.
- Aspect labels distinguish the object of a complaint.
- Topics summarize recurring term patterns under a model.

Rare Complaints Lower Precision



Synthetic classifier. Sensitivity is 80% and specificity is 90%.

Precision Depends on Prevalence

Let π be complaint prevalence, s sensitivity, and c specificity.

$$\Pr(Y = 1 \mid \widehat{Y} = 1) = \frac{s\pi}{s\pi + (1 - c)(1 - \pi)}$$

False positives can dominate a queue when the target condition is rare.

Expected Queue Composition

In 10,000 synthetic reviews, complaint prevalence is 1%.

```
n, prevalence = 10000, 0.01
sensitivity, specificity = 0.8, 0.9
tp = n * prevalence * sensitivity
fp = n * (1-prevalence) * (1-specificity)
precision = tp / (tp + fp)
assert round(precision, 4) == 0.0748
```

The expected queue contains 80 true complaints and 990 false positives.

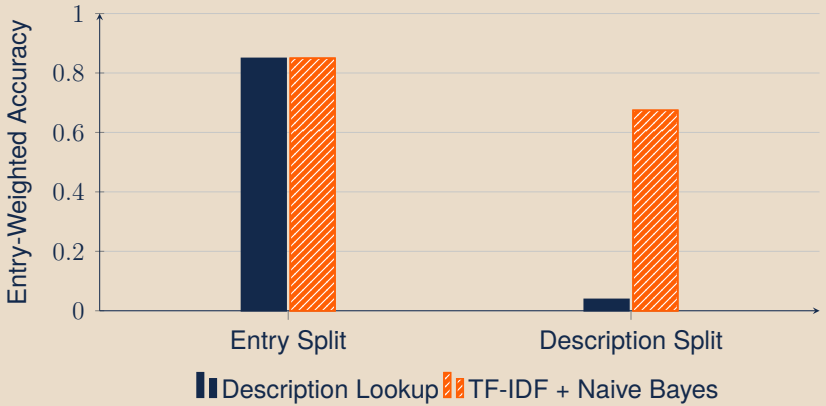
Human and LLM Annotation

- Write a label definition with difficult boundary examples.
- Measure agreement and adjudicate disagreements.
- Audit a held-out sample with human labels.
- Record the model, prompt, and response format.

mHealth Food Descriptions

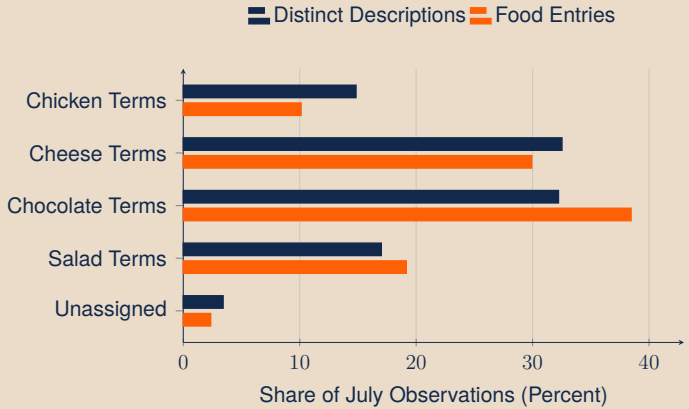
- Descriptions can recur across many users and dates.
- A repeated-description split evaluates familiar language.
- A held-out-description split tests generalization to new language.

Familiar and New Text Are Different Tests



mHealth text classification uses the stated training and evaluation splits.

Topic Shares Need a Denominator

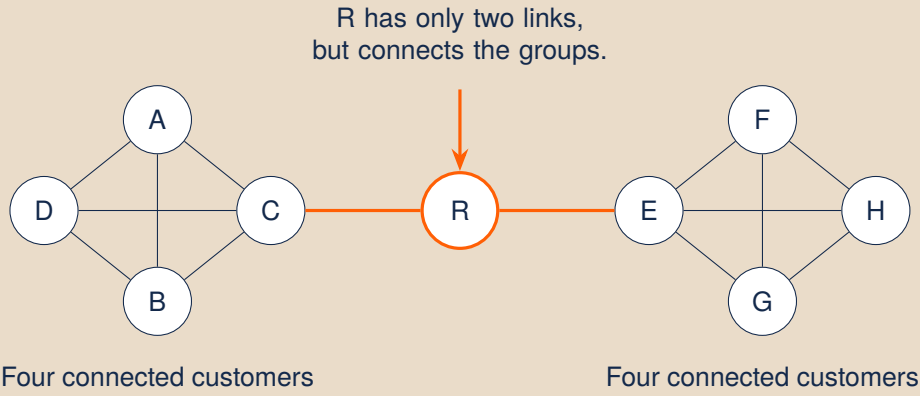


mHealth topic summaries compare documents, entries, and users under defined weights.

Network Objects

- Nodes identify customers or accounts.
- Edges represent a defined relationship in a time window.
- Degree counts links, while brokerage concerns paths between groups.

Few Links Can Connect Separate Groups



Synthetic undirected network. R has only two links.

Coding Exercise: A Label Audit

- Build a transparent keyword or TF-IDF baseline.
- Compare precision and recall on familiar and new descriptions.
- Inspect false positives and disagreements by label.

Engagement and Commercial Value

- Likes and comments measure platform responses.
- Sales and contribution require separate outcome records.
- Network similarity can reflect common interests or common exposure.

Authenticity of Reviews and Endorsements

- A generated product description and a claimed customer review serve different roles.
- A fabricated experience misrepresents the source of the evidence.
- Keep sponsored relationships and substantive editing traceable.

Text-Analysis Discussion

- Which label disagreement would change the service decision?
- How does the expected queue change when prevalence falls?
- What comparison would distinguish exposure from influence?

Marketing Analytics

Project Evidence and Midterm Presentations

By Nathan Yang

Session 08

A Decision and Its Evidence

- State the action a manager can change.
- Define the outcome, population, and horizon.
- Identify which evidence could change the proposed action.

A Traceable Data Table

- Name the original unit and the analysis unit.
- Reconcile rows and totals through joins and restrictions.
- Show one manually checked record or aggregate.

Evidence Categories

- Descriptive patterns summarize observed records.
- Prediction evaluates outcomes on held-out observations.
- Causal estimates require a justified counterfactual comparison.
- Scenarios calculate consequences under imposed inputs.

A Worked Presentation Case

- A paid-plan scenario has high predicted uptake.
- The underlying data contain food logging and no observed payments.
- The decision requires additional evidence on price, retention, and margin.

An Interpretable Figure

- Use a comparison that answers the stated question.
- Label units, denominator, and uncertainty.
- State whether plotted quantities are observed, estimated, or imposed.

A Reproducible Calculation

- Run from a clean session using documented inputs.
- Show a counterexample that checks a consequential choice.
- Keep complete code and results linked to the claim.

Presentation Discussion

- What is the strongest supported claim?
- Which assumption most affects the decision?
- What evidence should be collected next?

A Revision Decision

- Prioritize the error with the largest consequence for the action.
- Specify the additional data or comparison needed.
- Preserve a reproducible baseline for the revised analysis.

A Customer-Outcome Audit

- Identify who benefits and who can bear a cost.
- State whether the optimized metric can conflict with customer value.
- Name the relevant jurisdiction and verify the rule’s scope.

Sensitivity and Next Steps

- Report how the action changes across plausible inputs.
- Identify a decision threshold rather than only a point estimate.
- Select a feasible test that reduces consequential uncertainty.

Marketing Analytics

Experiments and Incremental Value

By Nathan Yang

Session 09



An Email-Campaign Decision

- Eligible customers receive either an email or no email.
- Measure purchases over a fixed response window.
- The decision concerns contribution per assigned customer.

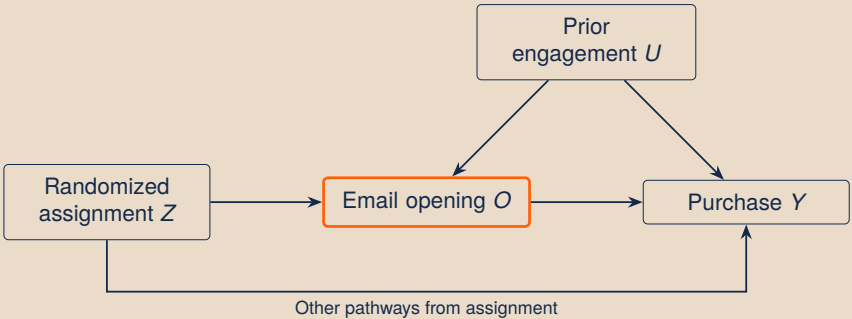
Potential Outcomes

For eligible customer *i*, define outcomes with and without assignment.

$$\tau = \mathbb{E}[Y_i(1) - Y_i(0)]$$

Each customer has one observed outcome under the realized assignment.

Selection Creates an Alternative Explanation



Opening depends on engagement and assignment.
Keep all assigned customers in the analysis.

Randomized assignment is distinct from engagement-driven email opening.

Email Assignment and Purchase

Assigned Group	Customers	Purchases	Rate
Email	1,000	130	13%
Control	1,000	100	10%

Synthetic experiment. Each customer has one binary outcome.

An Assignment-Effect Estimate

Compare purchase rates across all assigned customers.

$$\widehat{\tau} = \frac{130}{1000} - \frac{100}{1000} = 0.03$$

The estimated effect is three additional purchases per 100 assigned customers.

Sampling Uncertainty

Use an independent-group approximation for the synthetic binary outcomes.

$$\widehat{\mathrm{SE}} = \sqrt{\frac{.13(.87)}{1000} + \frac{.10(.90)}{1000}} \approx .0143$$

An approximate 95% interval is 0.21 to 5.79 percentage points.

Contribution and Decision Uncertainty

- An extra purchase contributes 40 dollars.
- Delivery costs 0.50 dollars per assigned customer.
- Estimated incremental contribution is 0.70 dollars per customer.

The Experiment Calculation

Reproduce the effect and its contribution implication.

```
from math import sqrt
p1, p0, n = 0.13, 0.10, 1000
effect = p1 - p0
se = sqrt(p1*(1-p1)/n + p0*(1-p0)/n)
interval = (effect - 1.96*se, effect + 1.96*se)
profit = 40 * effect - 0.50
assert round(profit, 2) == 0.70
```

The point estimate is positive while the contribution interval includes losses.

Power Begins With the Useful Effect

- Choose the smallest effect that could change the action.
- Use the expected outcome variance and assignment unit.
- Halving the target effect approximately quadruples sample size.

Implementation Checks

- Reconcile assigned counts and intended allocation.
- Check missing outcomes and tracking by assignment.
- Preserve the planned outcome horizon.
- Investigate exclusions that depend on treatment response.

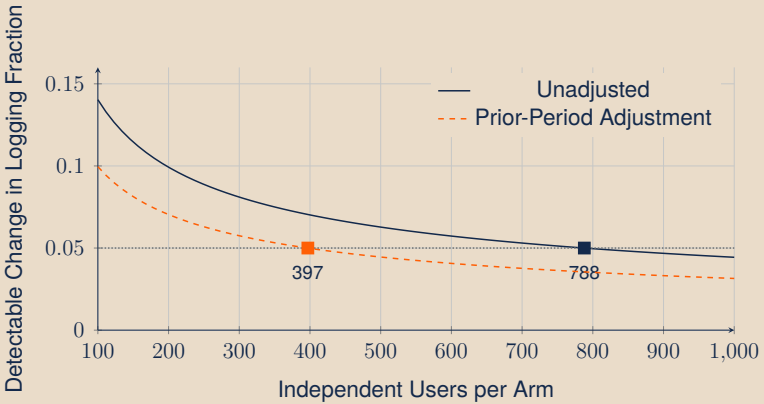
Prior Outcomes and Precision

Adjust the outcome using a pre-treatment measure X.

$$Y_i^{\text{adj}} = Y_i - \theta(X_i - \bar{X})$$

A predictive baseline can reduce residual variation while preserving the assignment comparison.

mHealth Baseline Activity Improves Precision



Planning calculation uses historical user-level variance and a five-point target.

Coding Exercise: An Experiment Plan

- Compute the synthetic email effect and interval.
- Change contribution, delivery cost, and the target effect.
- Specify an mHealth recommendation assignment with a fixed user denominator.

Interference and Assignment Units

- Customers can share offers or affect one another.
- Stores can divert purchases from nearby locations.
- The assignment unit should reflect the likely spillovers.

A Treatment Arm Needs a Defensible Design

- Random assignment identifies a comparison.
- The content of each treatment still affects the customers who receive it.
- Include customer outcomes, refusal, and stopping conditions.

Experiment Discussion

- Would a conversion gain cover the campaign cost?
- What changes if only email openers enter the analysis?
- Which guardrail outcome could reverse the rollout decision?

Marketing Analytics

Causal Comparisons and Policy Evaluation

By Nathan Yang

Session 10

A Repair-Policy Question

- A policy may change the cost of restoring restaurant availability.
- A national before-after change also includes common shocks.
- The target is the change attributable to the policy over a defined horizon.

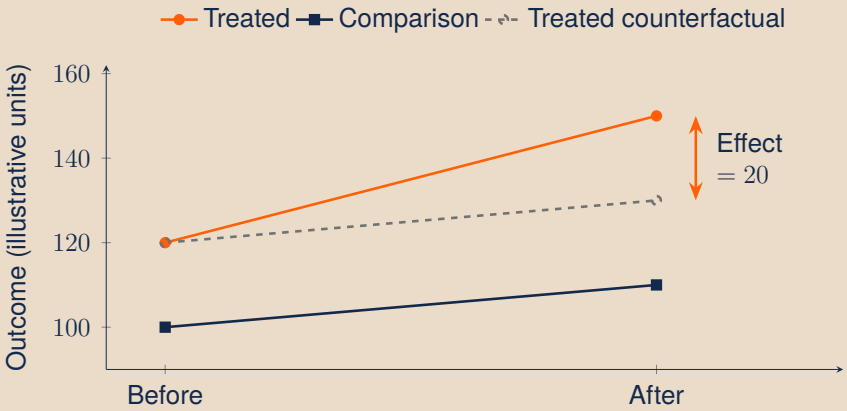
Difference-in-Differences

Subtract the comparison group's change from the treated group's change.

$$\widehat{\tau} = (\bar{Y}_{T,post} - \bar{Y}_{T,pre}) - (\bar{Y}_{C,post} - \bar{Y}_{C,pre})$$

Parallel untreated trends connect the observed changes to a causal effect.

The Counterfactual Trend

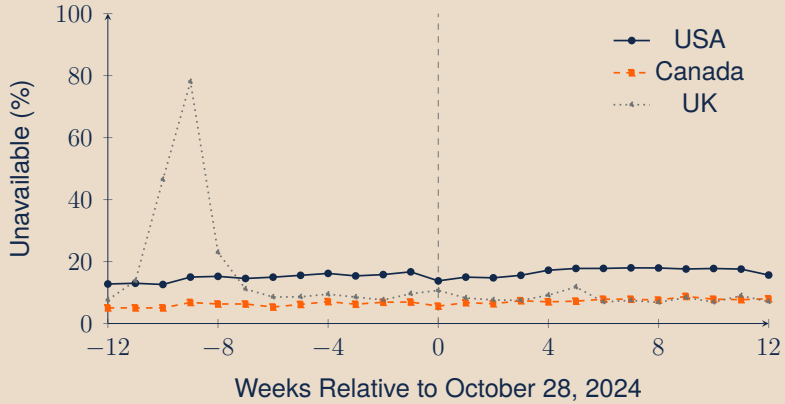


Synthetic outcomes. Treated sales rise 120 to 150 and comparison sales 100 to 110.

Design Checks

- Examine pre-treatment outcomes and institutional timing.
- Consider anticipation, spillovers, and changing composition.
- Match the comparison group to the policy's untreated counterfactual.

McBroken and the Comparison Group



A policy-analysis design requires a credible untreated comparison.

Regression Discontinuity

- Eligibility can change at a known score threshold.
- Nearby units require continuity of other outcome determinants.
- The resulting effect is local to the cutoff and compliance behavior.

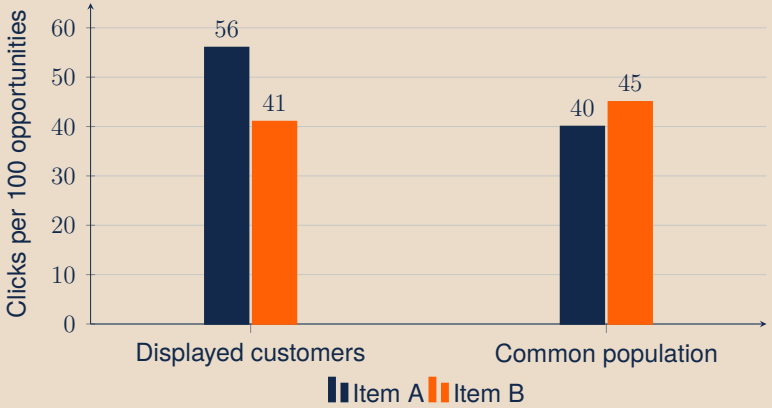
Instrumental Variables

- The instrument shifts treatment or price.
- Exclusion restricts its other paths to the outcome.
- Relevance alone does not establish exclusion.

Recommendation Logs Select Exposure

- Frequent buyers may usually see product A.
- Occasional buyers may usually see product B.
- Their raw click rates compare different customer populations.

Exposure Adjustment Reverses the Ranking



Synthetic randomized logging within frequent and occasional buyer groups.

Inverse-Propensity Policy Value

Weight each observed outcome by target-policy probability over logging probability.

$$\widehat{V}(\pi) = \frac{1}{n} \sum_i \frac{\pi(A_i \mid X_i)}{p_0(A_i \mid X_i)} Y_i$$

The denominator counts every eligible logged opportunity.

A Common-Population Calculation

Synthetic clicks and logging probabilities identify the two policy values.

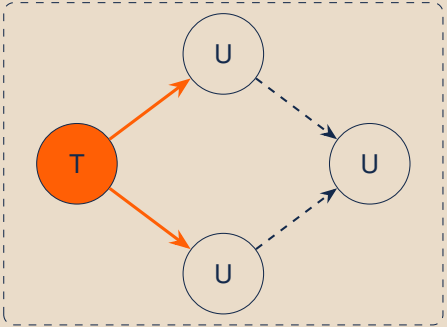
```
n = 1000
value_a = (270 / 0.90 + 10 / 0.10) / n
value_b = (25 / 0.10 + 180 / 0.90) / n
raw_a = 280 / 500
raw_b = 205 / 500
assert (value_a, value_b) == (0.40, 0.45)
```

The raw ranking and policy-value ranking disagree.

Support and Weight Concentration

- An action never shown to a group has no observed outcome there.
- Large weights amplify a small number of records.
- Clipping changes the estimator and its bias-variance tradeoff.

Spillovers Change the Outcome Comparison



One connected household or market can form a randomization cluster.

T: assigned treatment
U: assigned control

Control status does not guarantee absence of exposure through other people.

Synthetic store network with effects beyond assigned units.

Coding Exercise: Policy Value

- Reproduce raw rates and inverse-propensity values.
- Check them using within-group averages.
- Set one logging probability to zero and explain the missing comparison.

Information and Exposure

- A recommendation can draw attention to an item.
- A personalized endorsement can also change beliefs.
- Separate interventions can identify these channels.

A Policy Value for Whom?

- Clicks and sales can rise while customer value falls.
- A causal estimate concerns the outcome that was actually measured.
- Record refusal, completion, and customer time when procedures change.

Causal-Design Discussion

- What would the treated group have experienced without the policy?
- Which group or action lacks comparable support?
- What institutional evidence strengthens the proposed comparison?

Marketing Analytics

Prediction, Validation, and Marketing Actions

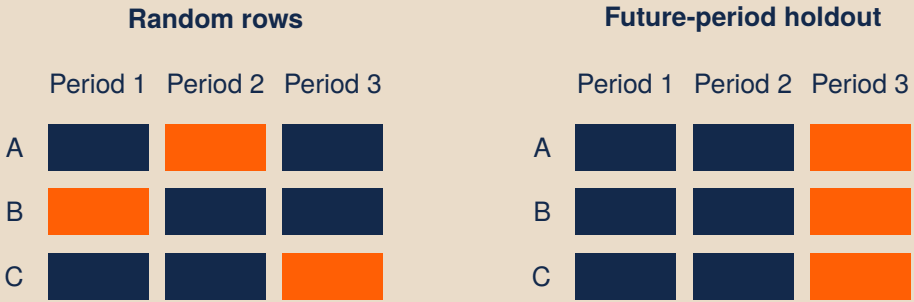
By Nathan Yang

Session 11

A Service-Capacity Decision

- A team can review 100 customers each week.
- Predictions use information available before that week.
- The outcome and review benefit determine the useful metric.

Validation Reproduces Deployment



Blue: training. Orange: test. Rows identify customers.
For new-customer deployment, hold out complete customer rows instead.

The deployment population determines the appropriate holdout.

The Forecast Origin

- Features end before the decision time.
- The outcome window begins after the decision.
- Training labels must already be observable at fitting time.

Baselines and Model Families

- Compare with a constant probability or recent-history forecast.
- Logistic regression supplies a probability baseline.
- Trees represent conditional splits. Ensembles combine trees.

Logistic Prediction

Transform a linear index into a binary-outcome probability.

$$\widehat{p}_i = \frac{1}{1 + \exp[-(\beta_0 + x_i^\top \beta)]}$$

A coefficient changes log odds. Its probability effect depends on the starting point.

Preprocessing Belongs in Training

- Fit imputation, scaling, and vocabulary on training records.
- Choose model settings using validation data.
- Keep the final test sample separate from model selection.

Rare Purchases and Accuracy

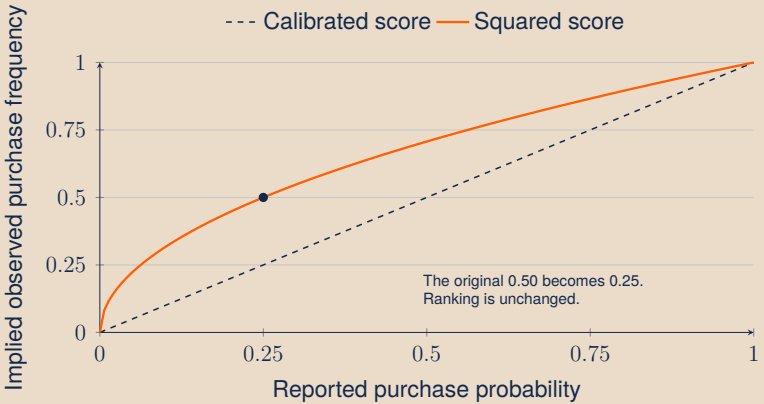
Quantity	Flagged	Not Flagged
Purchases	30	20
No Purchase	70	880

Synthetic population: 1,000 customers and 50 purchases.

Metrics and the Review Queue

- Precision: what fraction of reviewed cases are positive?
- Recall: what fraction of positives enter the queue?
- Calibration: do predicted probabilities match observed frequencies?

Ranking Can Stay Fixed While Probabilities Change



A synthetic transformation squares initially calibrated probabilities.

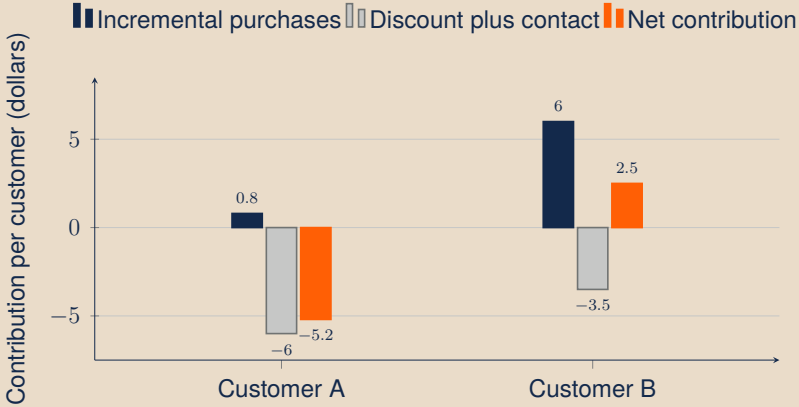
A Cost-Based Threshold

Suppose false positives cost C_{FP} and false negatives cost C_{FN}.

$$p > \frac{C_{FP}}{C_{FP} + C_{FN}}$$

With costs 1 and 9, the threshold is 0.10 under this simplified classification loss.

Coupon Targeting Includes the Untreated Outcome



Synthetic probabilities and costs. High treated response can have negative incremental value.

Queue Metrics

Compute metrics from the synthetic confusion matrix.

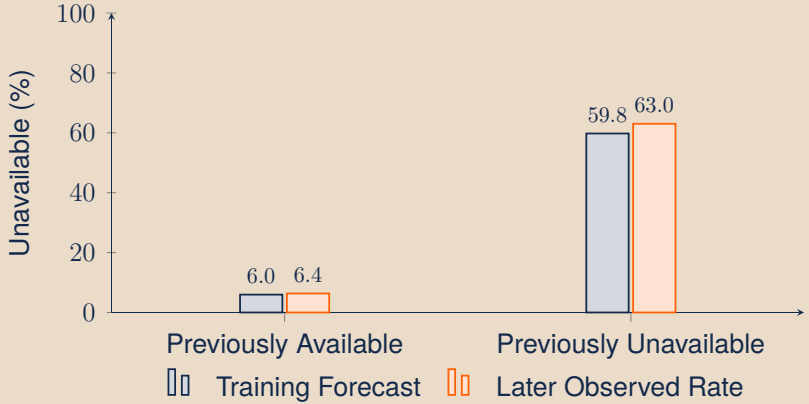
```
tp, fn, fp, tn = 30, 20, 70, 880
precision = tp / (tp + fp)
recall = tp / (tp + fn)
accuracy = (tp + tn) / (tp + fn + fp + tn)
assert (precision, recall, accuracy) == (0.3, 0.6, 0.91)
```

The model finds 60% of purchasers while 70% of its queue contains nonpurchasers.

McBroken Forecasting

- Predict availability on a later date.
- Retain a separate count of unknown future labels.
- Compare baseline and fitted forecasts on the same observed outcomes.

Prediction and Outcome Coverage



McBroken forecasts use later dates and observed-label denominators.

Coding Exercise: A Forecast Audit

- Create training, validation, and later test windows.
- Check every feature against the forecast date.
- Compare a simple baseline, ranking, and calibration.
- Report missing outcomes beside the test metrics.

Targeting and Delivered Opportunities

- An apparently neutral target can produce unequal delivery.
- Evaluate exposure and error costs across relevant groups.
- Housing, credit, and employment applications require their specific legal context.

Prediction Discussion

- Which error uses scarce review capacity?
- Which feature becomes available too late?
- Does the score support a forecast, an intervention, or both?

Marketing Analytics

Recommendation Systems and Customer Goals

By Nathan Yang

Session 12

A Returning-Customer Decision

- An mHealth user returns to log a meal.
- The system chooses among eligible categories using earlier histories.
- The objective may include familiar relevance and a stated variety goal.

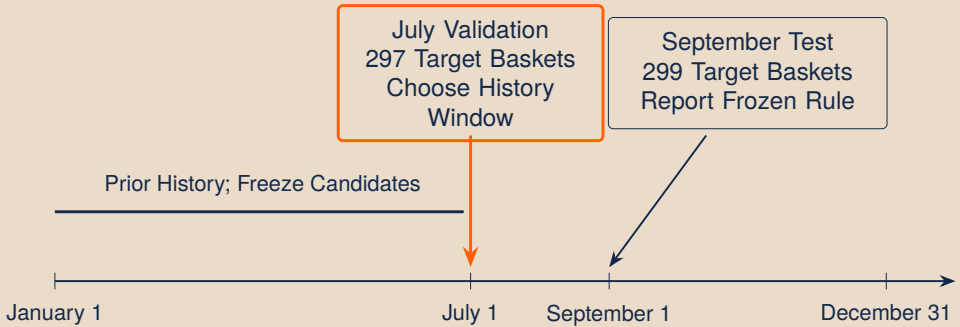
The Candidate Set

- Candidate generation determines what can be recommended.
- Scoring compares eligible items using current information.
- Ranking chooses order and applies list constraints.

Recommendation Models

- Content models compare item attributes and customer histories.
- Collaborative models learn patterns across customers and items.
- Popularity and personal-frequency rules provide useful baselines.

Histories Must Precede the Basket



All features precede each origin.
Each target is the first observed basket within that origin's month.

mHealth evaluation freezes histories and candidate sets at the forecast origin.

Recall Among Relevant Categories

Let L_i be the top-k list and B_i the categories in the next observed basket.

$$\text{Recall@k}_i = \frac{|L_i \cap B_i|}{|B_i|}$$

The denominator is the number of relevant basket categories for that user-event.

Position Matters

Discount lower-ranked relevant items when their positions affect exposure.

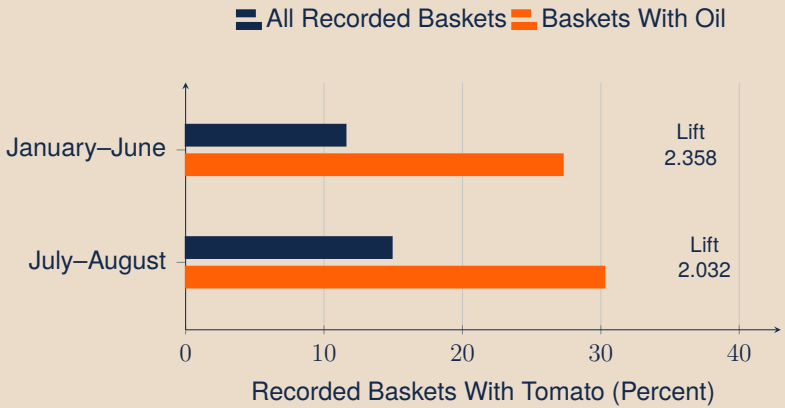
$$\mathrm{DCG@k} = \sum_{r=1}^{k} \frac{2^{\mathrm{rel}_r} - 1}{\log_2(r+1)}$$

NDCG divides by the best attainable DCG for the same relevance labels.

Basket Associations

- Support counts how often items occur together.
- Confidence conditions on the antecedent item.
- Lift compares confidence with the consequent’s marginal prevalence.

Lift Changes the Baseline Comparison



mHealth basket rules use distinct baskets and future validation.

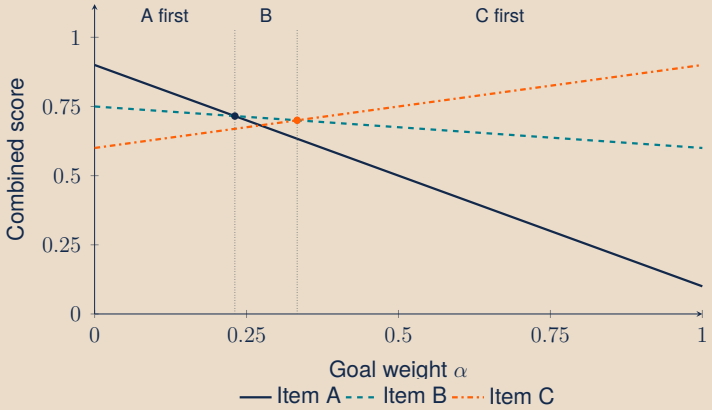
Relevance and a Customer Goal

Combine relevance s_j with a goal score g_j using an explicit weight.

$$v_j(\alpha) = (1 - \alpha)s_j + \alpha g_j$$

The scores need a comparable scale before the weight has a clear interpretation.

Small Weight Changes Can Change the First Item



Synthetic item scores. Relevance and goal scores stay fixed.

A Three-Item Basket Check

The list contains A, B, and C. The basket contains A and C.

```
from math import log2
ranked = ['A', 'B', 'C']
relevant = {'A', 'C'}
recall = len(set(ranked) & relevant) / len(relevant)
dcg = sum((x in relevant) / log2(r+2)
          for r,x in enumerate(ranked))
ideal = 1 + 1/log2(3)
assert recall == 1.0
```

Recall is 1.0, while NDCG is about 0.920 because B occupies the second position.

LLM Explanations

- Explain a recommendation using verified item attributes.
- Check whether stated reasons are factually supported.
- Evaluate explanation effects separately from ranking quality.

Offline and Online Outcomes

- A hit in a historical basket measures predictive overlap.
- A recommendation changes exposure and may change behavior.
- A randomized policy comparison measures the intervention outcome.

Coding Exercise: Ranking With a Goal

- Compare popularity and personal-frequency baselines.
- Tune the goal weight on validation baskets.
- Evaluate the fixed choice on later baskets.
- Audit one generated explanation against item attributes.

Customer Control of Recommendations

- A customer can accept a suggested goal or reject it.
- Factual explanations should distinguish evidence from an inferred preference.
- Evaluate whether the interface respects a revised goal.

Recommendation Discussion

- Which relevant item was excluded before scoring?
- Whose goal determines the ranking weight?
- What evidence would show improved customer experience?

Marketing Analytics

Adaptive Recommendations and Marketing Budgets

By Nathan Yang

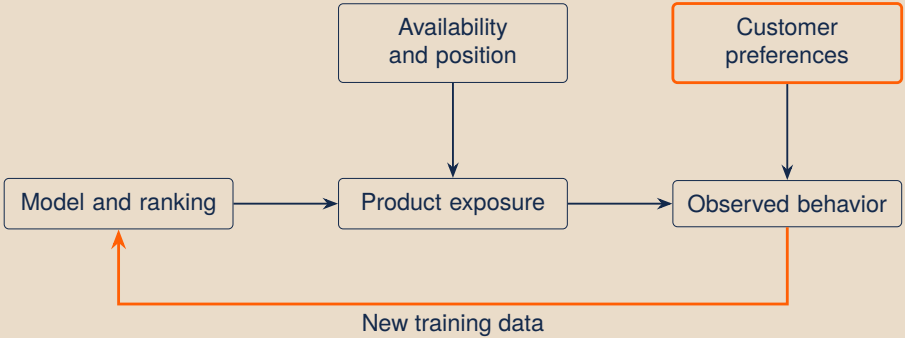
Session 13



An Adaptive Policy

- A model selects exposure using existing interaction histories.
- Customers respond to the displayed options.
- Retraining incorporates those selected responses.

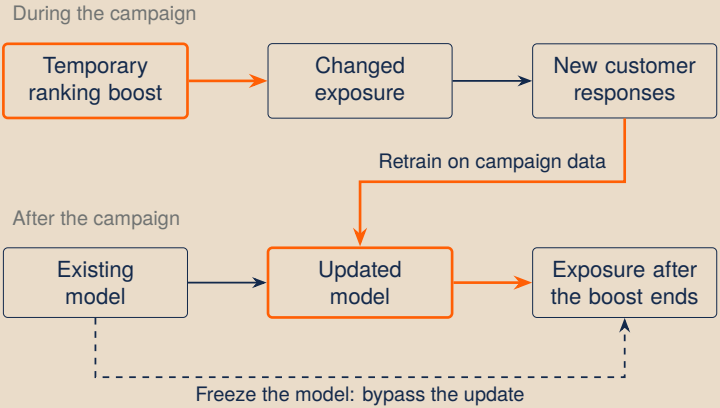
Selected Exposure Creates a Feedback Loop



Interaction data reflect both preference and opportunity.
A feedback loop can reinforce initially favored items.

The diagram describes a policy mechanism rather than an estimated effect.

The Evaluation Horizon Includes Learning



A temporary intervention can persist through subsequent model updates.

Customer Goals and Platform Objectives

- Clicks reveal a response within an offered environment.
- Considered preferences can differ from immediate engagement.
- Measure satisfaction, unwanted repetition, and customer-stated goals.

A List Objective

A list combines item relevance with category variety.

$$J(L) = \sum_{j \in L} s_j + \lambda K(L)$$

The variety weight λ is expressed in the same units as the relevance score.

Two Three-Item Lists

List	Relevance	Categories	Value at 0.30
A, B, C	2.50	2	3.10
A, C, D	2.30	3	3.20

Synthetic scores. The lists tie at a variety weight of 0.20.

Model-Based and Experimental Evidence

- A fitted choice model can compare alternative recommenders.
- A randomized perturbation can test predicted response patterns.
- Engagement, retention, revenue, and welfare are distinct outcomes.

Marketing-Mix Modeling

- Weekly channel spending can have delayed effects.
- Response can saturate at high exposure.
- Spending decisions can respond to anticipated demand.

Carryover

A geometric accumulated-exposure measure retains earlier spending.

$$a_t = s_t + \rho a_{t-1}, \qquad 0 \leq \rho < 1$$

A 100-unit pulse with $\rho = 0.5$ gives 100, 50, 25, and 12.5 in successive periods.

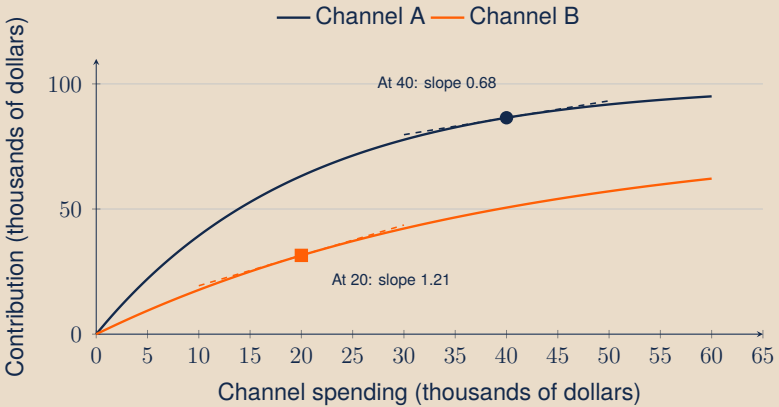
Saturation and the Next Dollar

A concave response has declining marginal contribution.

$$R(s) = A[1 - \exp(-s/b)], \qquad R'(s) = \frac{A}{b}\exp(-s/b)$$

Budget transfers depend on marginal response at the current allocation.

Average and Marginal Returns Can Disagree



Synthetic known response curves. Total spending is fixed at 60 thousand dollars.

The Fixed-Budget Optimum

Solve equal marginal returns for the imposed curves.

```
from math import exp, log
budget = 60
a = (log(2.5) + budget/40) / (1/20 + 1/40)
b = budget - a
ma = 5 * exp(-a/20)
mb = 2 * exp(-b/40)
assert abs(ma-mb) < 1e-12
```

The allocation is about 32.22 to A and 27.78 to B, in thousands of dollars.

Coding Exercise: A Policy Objective

- Enumerate three-item lists at several variety weights.
- Solve and independently grid-check the fixed-budget allocation.
- Report the assumptions that change the preferred action.

Attention and the Optimization Target

- A free service can require customer time and attention.
- More exposure can benefit advertisers while burdening customers.
- Evaluate voluntary use and customer outcomes alongside engagement.

Policy Discussion

- What changes when the recommender learns during the experiment?
- Which outcome measures the customer’s stated goal?
- How much uncertainty in response slopes would change the budget?

Marketing Analytics

Generative AI, Consumer Choice, and Regulation

By Nathan Yang

Session 14

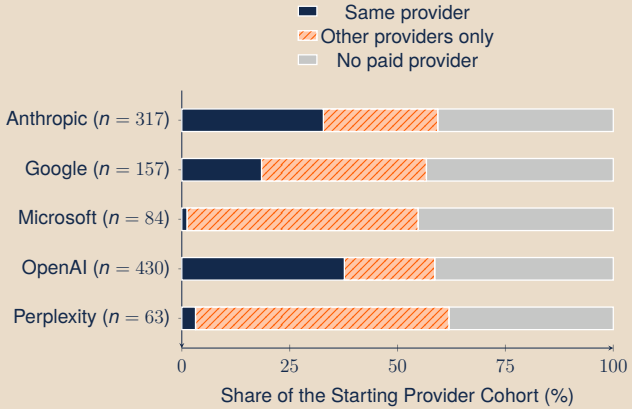
A Subscription-Growth Decision

- A firm wants to increase paid renewals.
- The evidence includes subscription histories, interfaces, and customer outcomes.
- A higher renewal rate can arise through improved value or greater cancellation friction.

Public Attention and Paid Demand

- Pageviews measure attention to selected pages.
- Paid subscriptions measure a specific payment relationship.
- User, provider, and category denominators differ.

A Fixed Starting Cohort

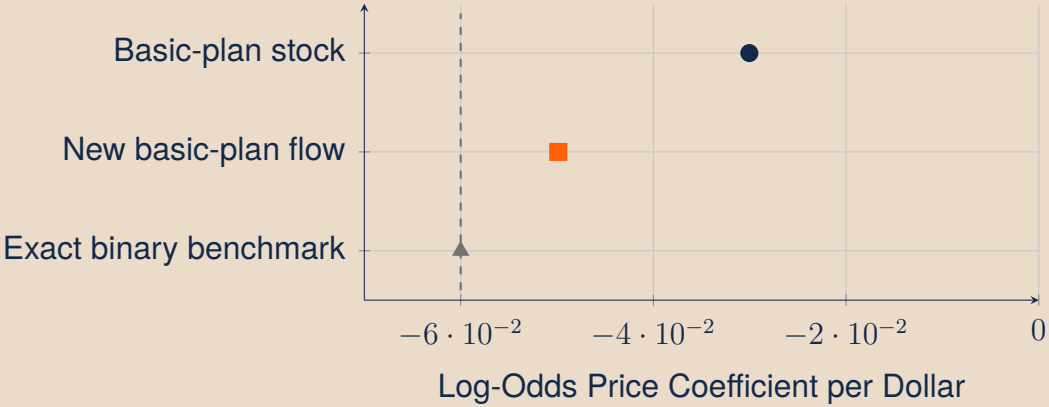


Entirely synthetic subscription panel. Provider labels identify simulated options.

Subscriber Stocks and New Flows

- A subscriber stock includes past acquisition and survival.
- A new-subscription flow can include first purchases and returns.
- A price coefficient depends on the outcome and behavioral model.

A Known Model Is a Useful Verification Case

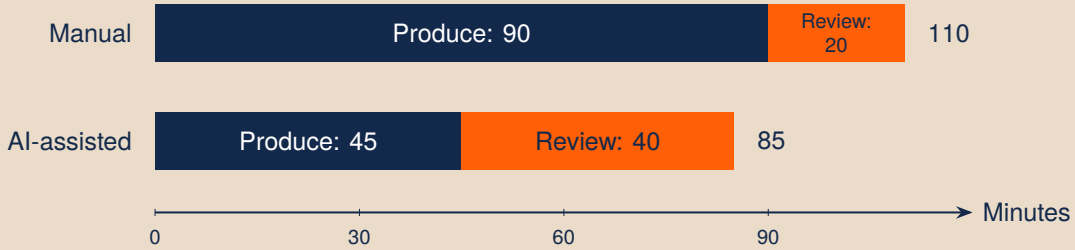


Synthetic demand exercise with a known binary-model benchmark.

AI Workflow Quality

- Measure the completed analysis and its validity.
- Include generation, review, and repair time.
- Use comparable tasks and independent quality criteria.

Review Time Is Part of Production



Drafting time falls 50%; total time falls 22.7%.
Task and completed-work quality held fixed.

Synthetic workflows meet the same stated quality requirement.

A Choice Procedure Can Change Renewal

- A direct offer and an offer preceded by questions can produce different choices.
- The customer still makes the final selection.
- An independent value measure determines whether an additional renewal benefits that customer.

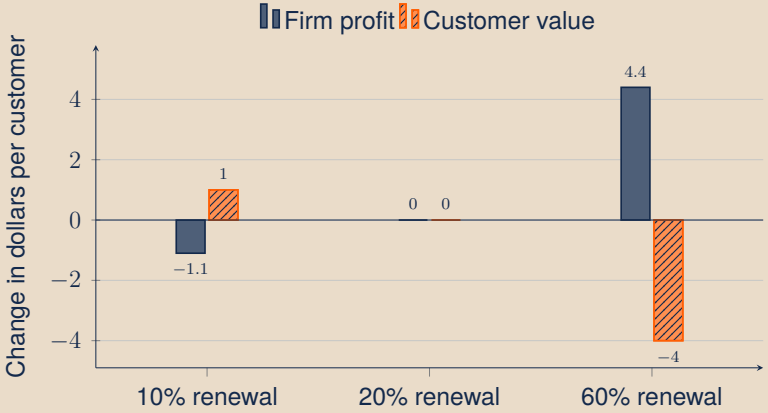
Opposed Interests

Suppose renewal yields firm margin $\nu > 0$ and costs the customer $\sigma > 0$ relative to declining.

$$\Delta\Pi = \nu(s - s_0), \qquad \Delta V = -\sigma(s - s_0)$$

A higher renewal probability raises profit and lowers customer value under these stated stakes.

Profit and Customer Value Can Move Oppositely



Synthetic probabilities and fixed stakes. Direct renewal is 20%.

Experimentation Selects the Commercial Winner

- A fixed set of procedures has stable renewal probabilities.
- Independent randomized samples estimate each probability.
- With opposed interests, accurately selecting the largest rate also selects the largest loss in that set.

A Cancellation Audit

- Include a direct-cancellation comparison.
- Record whether optional retention content can be refused before exposure.
- Measure completion, elapsed time, subsequent charges, and customer value.

Rule Status and Scope

- A statute, binding rule, interpretive guide, and proposal have different authority.
- Record jurisdiction, covered conduct, exceptions, and effective date.
- Use the current primary source before turning a result into a business recommendation.

Pricing, Reviews, and Delivery

- Price-display rules require attention to the covered industry.
- Reviews presented as customer experiences require truthful provenance.
- Housing-ad delivery can create discrimination concerns beyond targeting choices.

A Welfare Comparison

Synthetic stakes distinguish profit from customer value.

```
direct, selected = 0.20, 0.60
firm_margin, customer_loss = 11, 10
firm_gain = firm_margin * (selected-direct)
value_change = -customer_loss * (selected-direct)
assert round(firm_gain, 2) == 4.40
assert round(value_change, 2) == -4.00
```

Higher combined surplus can coexist with a customer loss under these assumptions.

Coding Exercise: A Marketing Audit

- Verify a demand denominator and the workflow-time calculation.
- Compute profit and customer value for alternative retention procedures.
- Propose an interface change and the evidence needed to evaluate it.

A Decision With Consumer Outcomes

- State the commercial target and customer outcome.
- Identify the procedure the customer can accept, refuse, or revise.
- Connect the proposed action to current rules and supporting evidence.

Marketing Analytics

Integrated Decisions and Final Presentations

By Nathan Yang

Session 15

An Integrated Marketing Decision

- Choose a feasible action and a defined customer population.
- Connect measurement, modeling, and evaluation to that action.
- State the costs, customer outcomes, and uncertainty that affect the choice.

The Evidence Behind the Recommendation

- State the sample, observation unit, and outcome horizon.
- Show the comparison that supports the main claim.
- Explain which result changes the recommended action.

A Calculation the Audience Can Check

- Trace one figure to an aggregate table and source records.
- Reproduce one prediction, coefficient interpretation, or policy value.
- Retain complete code and data definitions with the result.

A Result That Could Change the Decision

- Identify the most consequential uncertain parameter.
- Show the threshold at which the preferred action changes.
- Consider a feasible test that could resolve that uncertainty.

Customer Consequences

- Report effects on access, time, value, and unwanted charges where relevant.
- Distinguish voluntary continued use from difficulty leaving.
- Explain how customers can refuse or correct the proposed interaction.

Regulatory Scope

- Identify the jurisdiction and covered conduct.
- Separate the current rule from a proposal or allegation.
- Match the evidence supporting a claim to what customers will understand.

Presentation Discussion

- Is the recommendation supported for the stated population?
- What analytical or customer risk has the largest consequence?
- Which part of the result can be reproduced independently?

A Deployment Comparison

- Retain a benchmark and a defined evaluation population.
- Monitor outcomes and data coverage after the action changes behavior.
- Set a response when quality or customer outcomes deteriorate.

An Evidence-Based Recommendation

- Name the action, expected consequence, and decision threshold.
- Keep consequential uncertainty beside the estimate.
- Identify the next concrete extension supported by the evidence.